

Project Report 2

LLM-as-Judge Evaluation Pipeline

1. Project Title

LLM-as-Judge Evaluation Pipeline

2. Project Overview

The LLM-as-Judge Evaluation Pipeline is an AI evaluation framework that automatically assesses the quality of Large Language Model responses using another LLM. It supports pointwise scoring, pairwise comparison, bias analysis, structured validation, and automated report generation.

3. Objectives

- Evaluate LLM-generated responses.
 - Compare multiple responses automatically.
 - Generate structured evaluation reports.
 - Detect quality issues and response bias.
 - Build an AI quality assurance system.
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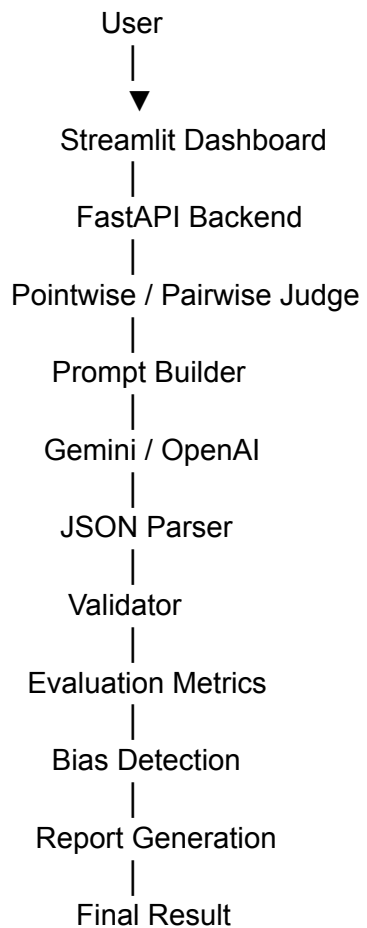
4. Problem Statement

Evaluating LLM responses manually is time-consuming and inconsistent. This project automates evaluation by using an LLM as a judge to provide objective scoring across multiple quality dimensions.

5. Tech Stack

- Python
 - FastAPI
 - Streamlit
 - Google Gemini
 - OpenAI
 - Pydantic
 - JSON Repair
 - Docker
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6. System Architecture



7. Workflow

1. User submits question and response(s).
 2. FastAPI receives request.
 3. Prompt is generated.
 4. Gemini/OpenAI evaluates response.
 5. JSON response is parsed.
 6. Schema validation is performed.
 7. Metrics are calculated.
 8. Report is generated.
 9. Results are displayed in Streamlit.
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8. Key Features

- Pointwise Evaluation
 - Pairwise Comparison
 - JSON Repair
 - Response Validation
 - Bias Detection
 - Interactive Dashboard
 - Automated Reports
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9. Evaluation Metrics

- Correctness
 - Relevance
 - Completeness
 - Fluency
 - Overall Score
 - Confidence
 - Reasoning
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10. API Endpoints

Endpoint	Description
/health	Health Check
/evaluate	Pointwise Evaluation

/pairwise Compare Two
Responses

11. Applications

- AI Chatbot Evaluation
 - RAG Evaluation
 - LLM Benchmarking
 - AI Quality Assurance
 - Research Evaluation
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12. Future Enhancements

- Multi-model benchmarking
 - Human evaluation integration
 - Leaderboards
 - Batch evaluation
 - Real-time dashboards
 - Explainable AI metrics
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13. Conclusion

The LLM-as-Judge Evaluation Pipeline provides an automated framework for evaluating and benchmarking AI-generated responses. It enables objective, scalable, and reproducible assessment of LLM quality, making it valuable for AI testing, chatbot evaluation, and model comparison.